

Final Project for DS535

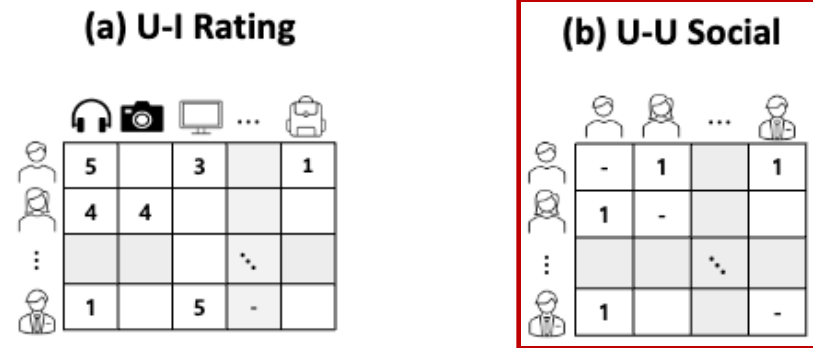
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Topic – Social Recommendation

- In recommender system, given data that represents interactions between users and items is too sparse.
- This sparsity of interaction data yields low-quality representations of users and items which are used for recommendation.
- To overcome this limitation, many approaches have **incorporated social graphs**, which leverage the connections between users to supplement the sparse user-item interaction data

→ **Social Recommendation (SR)**



Motivation – Social Network Analysis

There are some theories about social network:

- **Social Influence** → Users interaction behaviors are influenced by their friend, family, and others who are connected in social relations
- **Social Homophily** → Users tend to form social connection with individuals who share similar item interaction characteristics

→ Let's consider social connections by **assigning weights** from the perspectives of social influence

- Not all social edges will exert the same level of influence.

Idea – Soft-Denoising Social Network

Before reweighting, we need to determine **how we control social influence between user-user pair**

We suggest a proposition related to it

- *If users have bought **similar items** and they are socially connected, they **influence each other strongly***

→ This proposition is come from an assumption:

If users who are connected in the social network and each of them have interaction histories which include many similar items (**same category, same brand, same attribute, etc.**), they have influenced each other strongly

Method – Problem Definition

Input:

user set $U = \{u_1, \dots, u_m\}$

item set $I = \{i_1, \dots, i_n\}$

interaction matrix $\mathbf{R} \in \{0,1\}^{m \times n}$, $r_{ui} = 1$ if there is an interaction between user u and item i

social matrix $\mathbf{S} \in \{0,1\}^{m \times m}$, $s_{uv} = 1$ if user u is socially connected with user v

Design:

function $f: U \times I \rightarrow \mathbb{R}$ such that outputs the probability of interaction between a user and an item

Output:

prediction $\hat{y}_{i,j} = f(i, j)$ ($i = \text{user}, j = \text{item}$)

Method – Goal

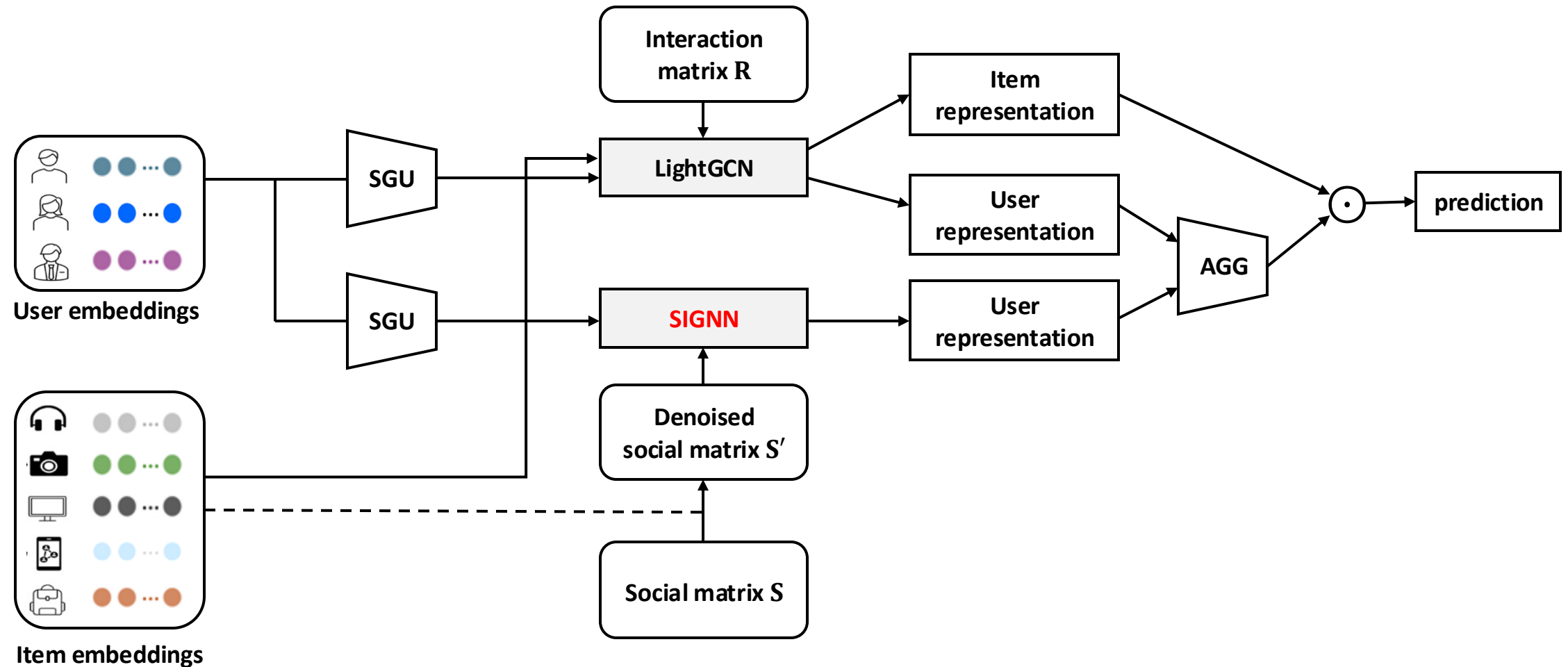
Reflect meaningful social influence into the representation of users based on **social influence control**

→ When designing f , let's use $\mathbf{S}' \in [0,1]^{m \times m}$, which integrates social influence into $\mathbf{S} \in \{0,1\}^{m \times m}$

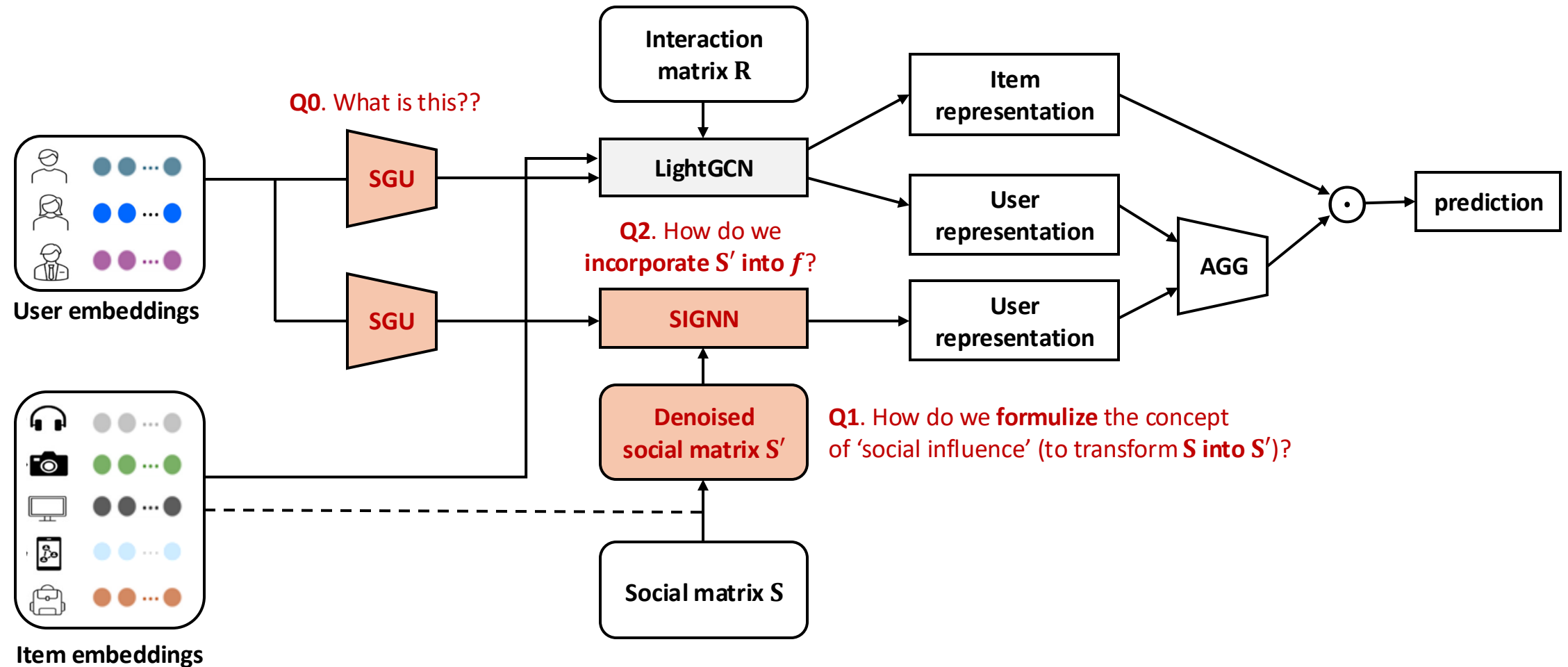
There are two remaining questions:

1. How do we **formulize** the concept of 'social influence' (to transform **S into S'**)?
2. How do we **incorporate S' into f**?

Method – Overall Framework



Method – Overall Framework



Method – Self Gating Units

Q0. What is SGU?

A0. The **self-gating units(SGU)** enables to control the information flow from the base user embeddings to each channel

To enable the user to capture different information from the interaction network and the social network, we formulate the equations as follows:

$$\mathbf{E}_u^{R(0)} = \text{SGU}_R \left(\mathbf{E}_u^{(0)} \right) = \mathbf{E}_u^{(0)} \odot \sigma \left(\mathbf{E}_u^{(0)} \mathbf{W}_R + \mathbf{b}_R \right)$$

$$\mathbf{E}_u^{S(0)} = \text{SGU}_S \left(\mathbf{E}_u^{(0)} \right) = \mathbf{E}_u^{(0)} \odot \sigma \left(\mathbf{E}_u^{(0)} \mathbf{W}_S + \mathbf{b}_S \right)$$

Method – Denoised Social Matrix

Q1. How do we formulize the concept of '**social influence**' (to transform S into S')?

A1-1. We set the embedding that represents **interaction behaviors** of user (e_u'):

- Proposition: *If users have bought similar items and they are socially connected, they influence each other strongly*

$$e_u' = \sum_{i \in N^R(u)} \frac{1}{\sqrt{d(u)}\sqrt{d(i)}} * e_i^{(0)}, \quad (d(\cdot) = \text{degree of node}, N^R(\cdot) = \text{neighbor set of node in } \mathbf{R})$$

A1-2. We calculate behavior similarity(=**social influence**) between **socially-connected** user pairs:

- Proposition: *If users have bought similar items and they are socially connected, they influence each other strongly*

$$s_{uv}' = \frac{1 + \text{sim}(e_u', e_v')}{2} \exp\left(-\frac{\|e_u' - e_v'\|_2^2}{2\sigma^2}\right)$$

Method – Social Influence based GNNs

Q2. How do we **incorporate S' into f** ?

A2. We follow **dual network** representation generation process

1) Interaction network: Applying LightGCN with SGU pre-filtered user embeddings (**not using S'**)

$$\mathbf{e}_i^{(l)} = \sum_{u \in N^R(i)} \frac{1}{\sqrt{d(i)}\sqrt{d(u)}} * \mathbf{e}_u^{R(l-1)}$$

$$\mathbf{e}_u^{R(l)} = \sum_{i \in N^R(u)} \frac{1}{\sqrt{d(u)}\sqrt{d(i)}} * \mathbf{e}_i^{(l-1)}$$

Method – Social Influence based GNNs

Q2. How do we **incorporate S' into f** ?

A2. We follow **dual network** representation generation process

2) Social network: Applying LightGCN with SGU pre-filtered user embeddings (**using S'**)

$$\mathbf{e}_u^{S(l+1)} = \sum_{v \in N^S(u)} \frac{\mathbf{S}'_{uv}}{\sqrt{d(u)}\sqrt{d(v)}} \mathbf{e}_v^{S(l)}$$

Limitation – Even if a user has a significant influence on their surrounding peers, the effect remains minimal due to the normalization factor ($\sqrt{d(u)}\sqrt{d(v)}$)

Proposition – The degree of social influence propagation in a social network should be proportional to the **importance** of the node within the entire network.

Method – Social Influence based GNNs

Q2. How do we **incorporate S' into f** ?

A2. We follow **dual network** representation generation process

2) **Social network**: Applying **Social Influence based GNNs(SIGNN)** with SGU pre-filtered user embeddings (**using S'**)

Q2-1. What is **SIGNN**?

A2-1. To account for the strength of social influence **between users** and a user's importance (the user's influence **in the network**), we propose a new normalization term during message aggregation leveraging the PageRank algorithm

$$\mathbf{e}_u^{S(l+1)} = \sum_{v \in N^S(u)} \mathbf{S}'_{uv} * \sqrt{\text{PageRank}(u)} \sqrt{\text{PageRank}(v)} \mathbf{e}_v^{S(l)}$$

Method – Prediction

We obtain the user and item representations by summing up all the layer-wise embeddings on each network:

$$\mathbf{E}_u^R = \sum_{l=0}^L \mathbf{E}_u^{R(l)}, \mathbf{E}_i = \sum_{l=0}^L \mathbf{E}_i^{(l)}, \quad \mathbf{E}_u^S = \sum_{l=0}^L \mathbf{E}_u^{S(l)}$$

Also, we use a convex summation to merge the user's representations:

$$\mathbf{E}_u = \alpha \mathbf{E}_u^R + (1 - \alpha) \mathbf{E}_u^S$$

Then, we evaluate the predicted score by

$$\hat{y}_{u,i} = f(u,i) = \mathbf{e}_u^T \mathbf{e}_i$$

Method – Training

We use **BPR(Bayesian Personalized Ranking) loss** for model training

$$\mathcal{L}_{BPR} = - \sum_{(u,i,j) \in \mathcal{D}} \log \sigma(\hat{y}_{u,i} - \hat{y}_{u,j})$$

Also, we add **contrastive loss** as regularizer term (**for generating better embeddings**)

$$\mathcal{L}_{CL}^{user} = - \sum_{u \in U} \log \frac{\exp(\text{sim}(\mathbf{e}_u, \mathbf{e}_u))}{\sum_{v \in U} \exp(\text{sim}(\mathbf{e}_u, \mathbf{e}_v))}, \quad \mathcal{L}_{CL}^{item} = - \sum_{i \in I} \log \frac{\exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_i))}{\sum_{j \in I} \exp(\text{sim}(\mathbf{e}_i, \mathbf{e}_j))}$$

If we define $\mathcal{L}_{CL} = \mathcal{L}_{CL}^{user} + \mathcal{L}_{CL}^{item}$, the final objective becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{BPR} + \lambda_1 \mathcal{L}_{CL} + \lambda_2 ||\boldsymbol{\theta}||^2$$

Experimental Datasets

We conducted experiments on three datasets containing user-item interactions and trust relations among users in the domains of business, shopping, and music

- Both interaction density and relation density are **extremely sparse**

Dataset	Yelp	LastFM	Ciao
# of Users	43,043	1,892	7,375
# of Items	66,576	17,632	105,114
# of Interactions	283,512	92,834	284,086
Interaction Density	0.01%	0.28%	0.04%
# of Relations	506,408	25,434	57,544
Relation Density	0.02%	0.71%	0.11%

Experimental Setup

- **Baselines:** LightGCN, LightGCN+Social, MHCCN, DSL
- **Evaluation metrics:** {Recall, NDCG}@{10, 20, 40}
- **Hyperparameter tuning:** Grid search
 - The number of layers l
 - Contrastive loss weight λ_1
 - Regularization loss weight λ_2
 - Proportion of interaction-aware user embedding α

Overall Performance Comparison

It does not outperform all baselines... 😞

However, it demonstrates generally **stable** performance and significantly surpasses the latest SR model.

The simple propagation process, which adjusts social influence and considers importance within the social network, proves to be an **easy yet effective** approach

Method	Yelp						LastFM						Ciao					
	R@10	R@20	R@40	N@10	N@20	N@40	R@10	R@20	R@40	N@10	N@20	N@40	R@10	R@20	R@40	N@10	N@20	N@40
LightGCN	0.0339	0.0534	0.0853	0.0169	0.0218	0.0283	0.1917	0.2682	0.3631	0.2429	0.2691	0.3111	<u>0.0399</u>	0.0612	0.0867	<u>0.0432</u>	<u>0.0484</u>	<u>0.0555</u>
LightGCN+Social	0.0357	0.0565	0.0893	0.0179	0.0231	0.0298	0.1893	0.2625	0.3569	0.2398	0.2641	0.3058	0.0373	0.0577	0.0817	0.0403	0.0453	0.0521
MHCN	0.0337	0.0547	<u>0.0876</u>	0.0168	0.0220	0.0287	0.2066	0.2866	0.3778	0.2614	0.2880	0.3284	0.0408	<u>0.0610</u>	<u>0.0868</u>	0.0443	0.0487	0.0559
DSL	0.0337	0.0544	0.0875	0.0171	0.0223	0.0291	0.1963	0.2719	<u>0.3702</u>	0.2488	0.2738	0.3172	0.0396	0.0575	0.0837	0.0422	0.0462	0.0536
IDEA	<u>0.0344</u>	<u>0.0553</u>	0.0874	<u>0.0175</u>	<u>0.0227</u>	<u>0.0293</u>	<u>0.1968</u>	<u>0.2741</u>	0.3701	<u>0.2504</u>	<u>0.2759</u>	<u>0.3183</u>	0.0392	0.0605	0.0871	0.0423	0.0473	0.0549

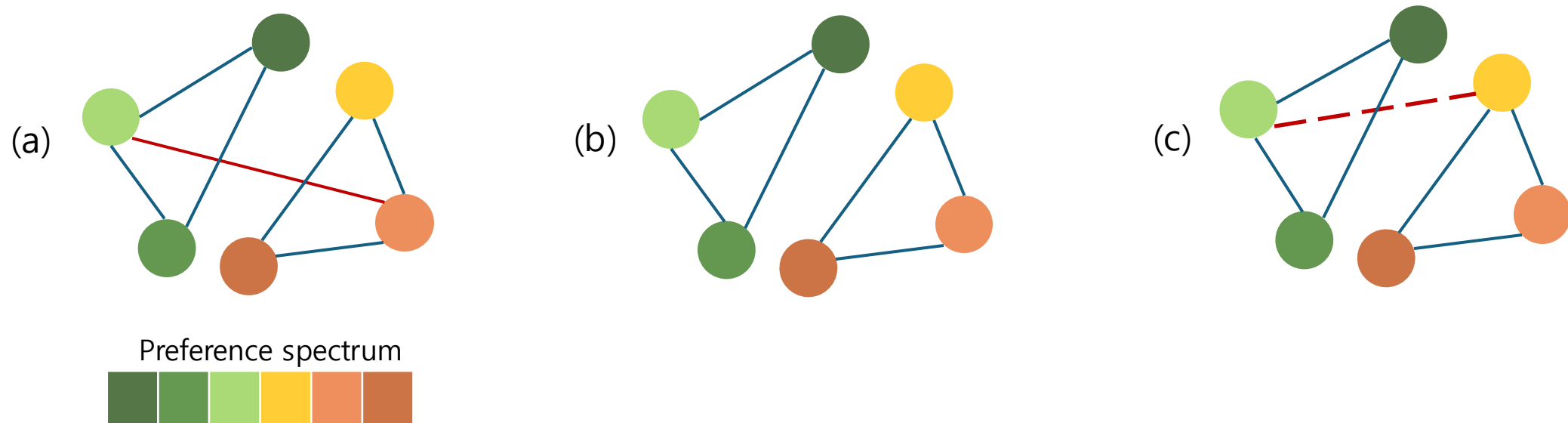
Another Approach – Rerouting Social Network

Q) How to further improve the masking process?

When removing the connections between low homophily users in the social graph,

- Existing approaches may remove relevant connections for future predictions
- Communities might get disconnected

Rerouting approach: Replace an edge between two **heterophilous** users with edges connecting their neighborhoods, prioritizing **homophilous** connections



Another Approach – Rerouting Social Network

Rerouting Algorithm

Algorithm 1 Rerouting Through a Mutual Friend

```
1: Input: Graph  $G = (V, E)$ , Nodes  $A, B$ , Similarity function  $\text{similarity}(X, Y)$ 
2: Output: Modified graph  $G'$ 
3: Compute  $a = \arg \max_{x \in \{A \cup \text{Neighbors}(A)\}} \text{similarity}(x, B)$ 
4: Compute  $b = \arg \max_{y \in \{B \cup \text{Neighbors}(B)\}} \text{similarity}(y, A)$ 
5: if  $\text{similarity}(A, b) > \text{similarity}(B, a)$  then
6:   Set  $C = b$ 
7: else
8:   Set  $C = a$ 
9: end if
10: Modify  $G$  by rerouting edge  $(A, B)$  through  $C$ 
11: Return: Modified graph  $G'$ 
```

Experimental Result

- Evaluation results on the FM dataset

Method	R@10	R@20	R@40
Idea	0.1968	0.2741	0.3701
Rerouting	0.1854	0.2615	0.3529

Conclusion

- We designed a model suited to social networks by adjusting the weights based on social influence and incorporating the importance of nodes during the message propagation process in the GNN.
- We experimented with various variants that integrate the social network and the interaction network.

Future Work:

- Explore other properties of social networks, such as social homophily, to refine the model structure.
- Experiment with alternative metrics instead of PageRank or incrementally modify the modules, such as using an MLP for fusing user representations during aggregation.

Thank you